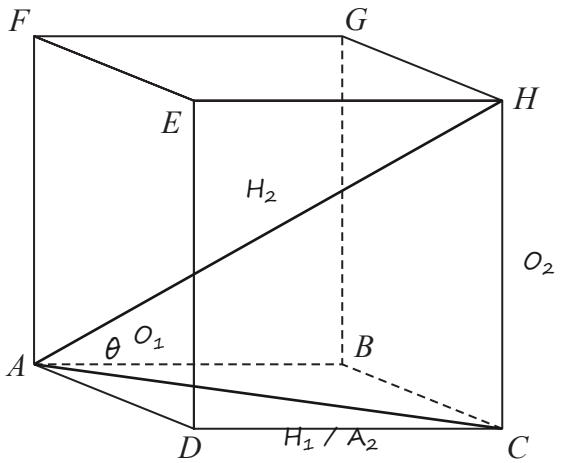
**18** *ABCDEFGH* is a cuboid.




$AB = 7.3$ cm
$CH = 8.1$ cm
Angle $BCA = 48°$

Find the size of the angle between $AH$ and the plane $ABCD$.
Give your answer correct to 1 decimal place.

θ is angle CAH: AC is the projection of AH onto the base.

Triangle ABC: $O_1 = AB = 7.3$, $H_1 = AC$.

$$\sin 48° = \frac{7.3}{AC}$$

**1 mark**
Use the correct base triangle

$$AC = \frac{7.3}{\sin 48°} \approx 9.823118926 \text{ cm}$$

**1 mark**
Find the base diagonal

Triangle ACH: $O_2 = CH = 8.1$, $A_2 = AC$.

$$\tan \theta = \frac{8.1}{AC}$$

**1 mark**
Use opposite ÷ adjacent for θ

$\theta = \tan^{-1}(8.1 \div 9.823118926...) = 39.5084923...°$

Keep the unrounded AC in the calculator.

**1 mark**
Round the angle to one decimal place

39.5 °

**(Total for Question 18 is 4 marks)**